

Project Proposal

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Project Title: Modeling light pollution using various geospatial data sources

Description of Data/Problem: Light pollution is a night-time phenomenon caused by the atmospheric scattering of excess light from things such as street lamps or other outdoor light fixtures that then bounces off of molecules in the air. Light pollution is commonly measured as night sky brightness (in magnitudes per square arcsecond), but is typically only directly measured at sites like observatories. Even then, data from these observatories is rarely accessible in real-time. The Globe at Night campaign created a monitoring network for observatories across the world to provide night sky brightness data for public use, with data going back as far as 2014. Light pollution, being an atmospheric phenomenon, is heavily impacted by weather conditions. The objective for this project is to predict light pollution (night sky brightness) at a given location using primarily meteorological, energy usage, and population data.

Proposed Solution: Using supervised learning techniques, associate the light pollution data for observatories to their distances from population centers, local energy usage, and meteorological conditions. We plan to use the processed data to construct a regression model to predict light pollution for any location.

Data Sources:

1. Light pollution data (night sky brightness)
<https://globeatnight.org/gan-mn/>
2. Meteorological data (multiple variables):
<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels>
3. U.S. Energy Information Administration Open Data (Maybe draw a correlation to state energy usage?)
<https://www.eia.gov/opensdata/>
4. International Energy Agency (country-level data on energy topics)
<https://www.iea.org/data-and-statistics/data-sets?filter=free>
5. US Population Data
<https://www.census.gov/data/datasets.html>

6. Population Weighted Centroid Positions by Zip Code (US only)

<https://catalog.data.gov/dataset/zip-code-population-weighted-centroids>

7. Census International Database Subnational Data

<https://www.census.gov/data-tools/demo/data/idb/excel/IDBSubnational.zip>